Homework

We consider a short position in a put option with strike κ and maturity T, delta-hedged using the underlying stock. The stock price dynamics are given by

$$\frac{dS_t}{S_t} = \mu\, dt + \sigma\, dW_t,$$

where μ and $\sigma > 0$ are constants, W is a Brownian motion under the physical measure $\mathbb{P}$, and $r > 0$ is the constant risk-free rate.

At time t, suppose your account has a short position of 1 put option and $Y_t = 0$ dollars in the money market. You decided to delta-hedge the put option using the underlying stock, so the position in the stock is

$$h_t = \partial_s P^{BS}(t, S_t),$$

where $P^{BS}(t, s)$ denotes the Black–Scholes call price evaluated at $S_t = s$. Accordingly, the amoount in the money market is

$$Y_t^{\text{new}} = Y_t - h_t S_t,$$

and the portfolio value is therefore

$$V_t = Y_t^{\text{new}} + h_t S_t - P^{BS}(t, S_t).$$

Since h_t remains constant over $[t, t + \Delta]$ and the money market position grows at the risk-free rate r, the end-of-day portfolio value is

$$V_{t+\Delta} = h_t S_{t+\Delta} + Y_t^{\text{new}} e^{r\Delta} - P^{BS}(t + \Delta, S_{t+\Delta}).$$

Our goal is to estimate the 10-day VaR of the portfolio over the period $[t, t + J\Delta]$ with $J = 10$ and $\Delta = 1/252$. The following parameters are given:

- $\alpha = 0.95$, $t = 0$, $\Delta = 1/252$, $T = 0.292$, $S_t = 152.51$, $\kappa = 170$, $r = 0.119\%$, $\sigma = 49.07\%$, and $\mu = 16.91\%$.

Perform the following tasks.

1. For a short position of 100 puts, use a simulation with 40,000 runs to compute the one-day VaR over $[t, t + \Delta]$. Then estimate the 10-day VaR by applying the square-root-of-time rule, namely by multiplying the one-day VaR by $\sqrt{10}$.

2. Estimate the $K = 10$ day VaR over $[t, t+K\Delta]$ using the time-aggregation procedure described above.

3. Compare the above two resulting VaR estimates obtained previously. Discuss any differences and explain the likely reason for the difference.